AP PRECALCULUS

Unit 1 • Worksheet 1.1A

Worksheet 1.1A • Change in Tandem

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

The table gives values of f. Which statement is consistent with the table?

x	0	2	4	6	8
$f(x)$	3	8	11	12	12.5

(A) f is increasing and concave up.

(B) f is increasing and concave down.

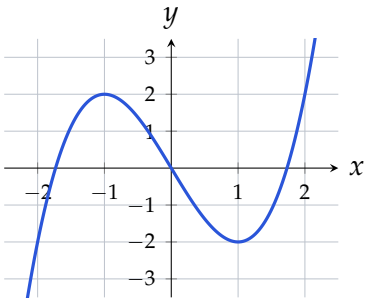
(C) f is decreasing and concave up.

(D) f is decreasing and concave down.

Question 2

NO CALCULATOR

The graph of g is shown. On which interval is g decreasing and concave up?



(A) $(-1, 0)$

(B) $(0, 1)$

(C) $(1, 2)$

(D) $(-2, -1)$

Question 3

NO CALCULATOR

Which statement must be true if the rate of change of h is negative and increasing on an interval?

(A) h is increasing and concave down.
(B) h is decreasing and concave down.
(C) h is decreasing and concave up.
(D) h is increasing and concave up.

Question 4

NO CALCULATOR

What are the zeros of $p(x) = x(x-3)(2x+5)$?

(A) $0,\ 3,\ 5$
(B) $0,\ -3,\ \frac{5}{2}$
(C) $3,\ -\frac{5}{2}$
(D) $0,\ 3,\ -\frac{5}{2}$

Question 5 (Free response)

NO CALCULATOR

A tank is filling with water. The volume $V(t)$, in liters, is recorded every 10 minutes.

t (min)	0	10	20	30	40
$V(t)$ (L)	40	100	150	190	220

(a) Is V increasing or decreasing on $[0, 40]$? Explain.
(b) Find the change in volume over each 10-minute interval. Is the graph of V concave up or concave down? Explain.
(c) What does your answer to (b) say about how fast the tank is filling?

Question 6 (Free response)

NO CALCULATOR

Sketch a graph of a function that is increasing on $(-\infty, 2)$, decreasing on $(2, \infty)$, concave down everywhere, and has zeros at $x = -1$ and $x = 5$.